

Knowledge Based Engineering (KBE) AE4204

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Lecture 8

*KBE opportunities in Engineering Design and
MDO
The KBE life-cycle*



Contents of this lecture

- KBE benefit in product development
- When to apply KBE?
- KBE to support MDO
 - The Design and Engineering Engine framework
 - The Multi Model Generator
 - High Level Primitives & Capability Modules
 - Examples past and recent projects
- The KBE life cycle (development methodology)
- AE4205+AE4204 in 3 bullet points

KBE benefits in product development

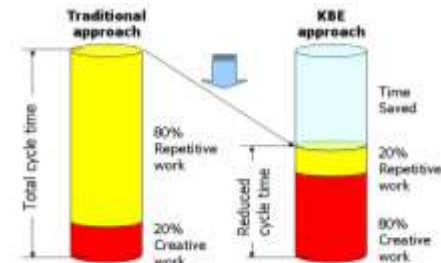
KBE benefit in product development

- Enhanced **productivity** : more designs generated and analysed in less time
- Improvement **quality** : more mature designs due to **improved design space exploration, more tradeoff studies** and guaranteed compliance to standards and best practices
- Short lead time** and **improved confidence** in generated designs are beneficial to generate (more) **competitive proposals**
- Improved level of *workers' satisfaction* and supported *innovation*: **less repetitive work, more time for creativity**
- Corporate **knowledge retention**

MORE

BETTER

FASTER

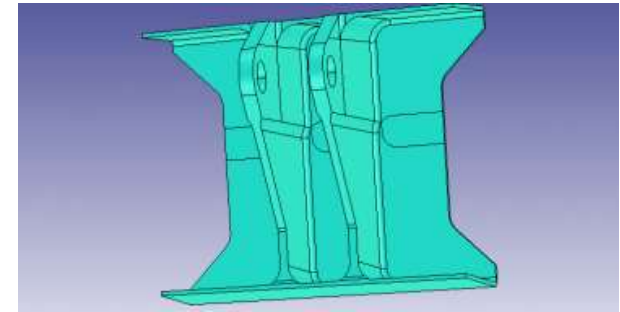


When to apply KBE?

Identification of proper KBE application cases

KBE is most effective:

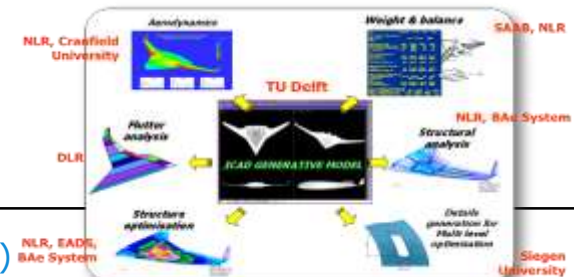
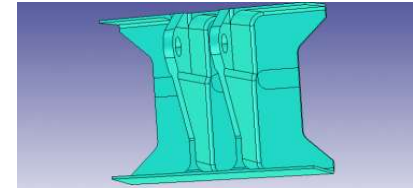
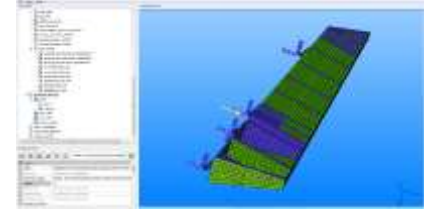
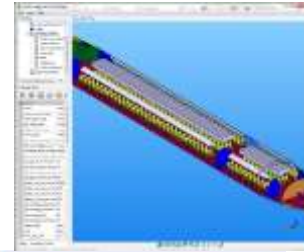
- In **conceptual design**, when configuration changes are frequent and significant (architectures tradeoffs)
- In **detailed design**, when product configuration is frozen and even small last-minute changes require extensive manual rework



Identification of proper KBE application cases

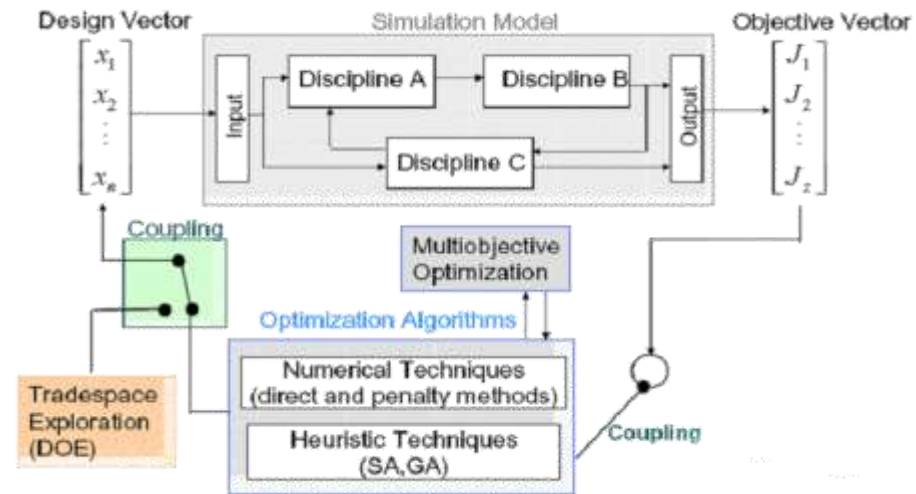
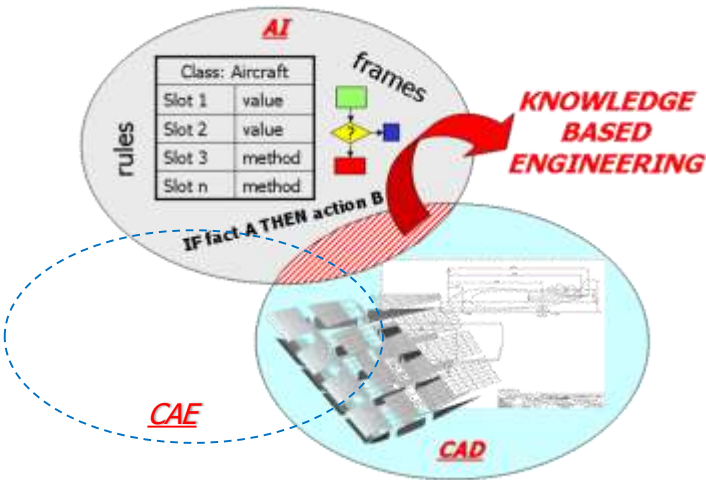
KBE is most effective when:

- The design case is **highly rule-driven**
- The **design process is well understood** and **consolidated**
- The design process is **repetitive and iterative**
- The problem is **multidisciplinary**
- The process involves a combination of:
 - **geometry manipulation**
 - **data processing**
 - **engineering decision-making**



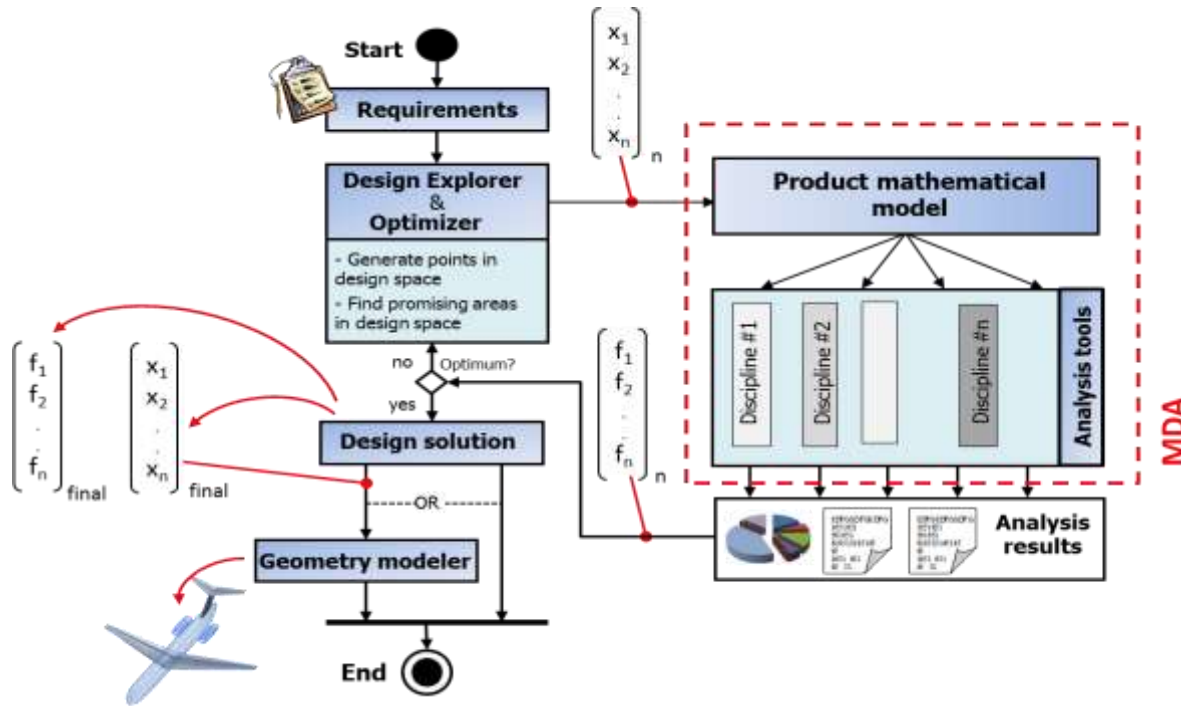
KBE to support MDO

KBE-MDO: *what was the link again?*



Traditional MDO approaches

MDO system **without** geometry



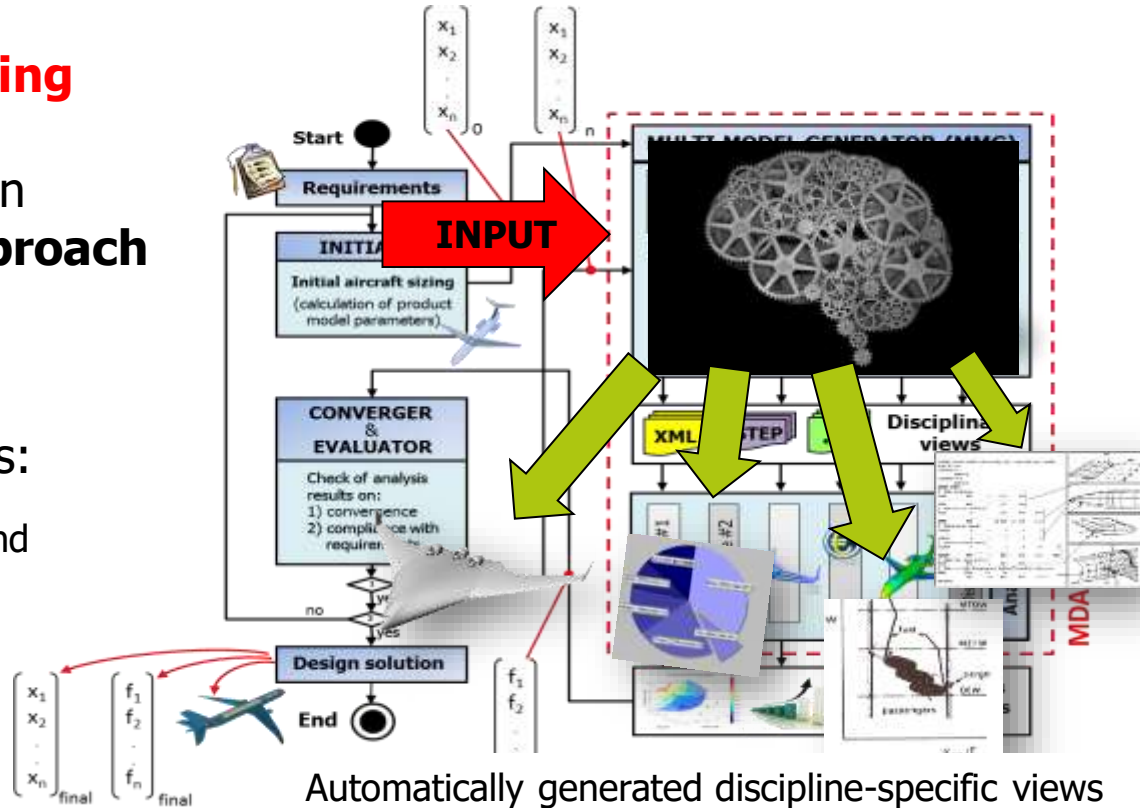
- No high fidelity
- No suitable for novel configurations

KBE-enabled MDO framework

The **Design and Engineering Engine (DEE)**:
an MDO framework based on
generative geometry approach

The **KBE Multi Model Generator (MMG)** provides:

- CAD-grade geometry generation and advanced product configuration
- fully automatic, robust model generation
- capture and re-use of *product* and *process* knowledge

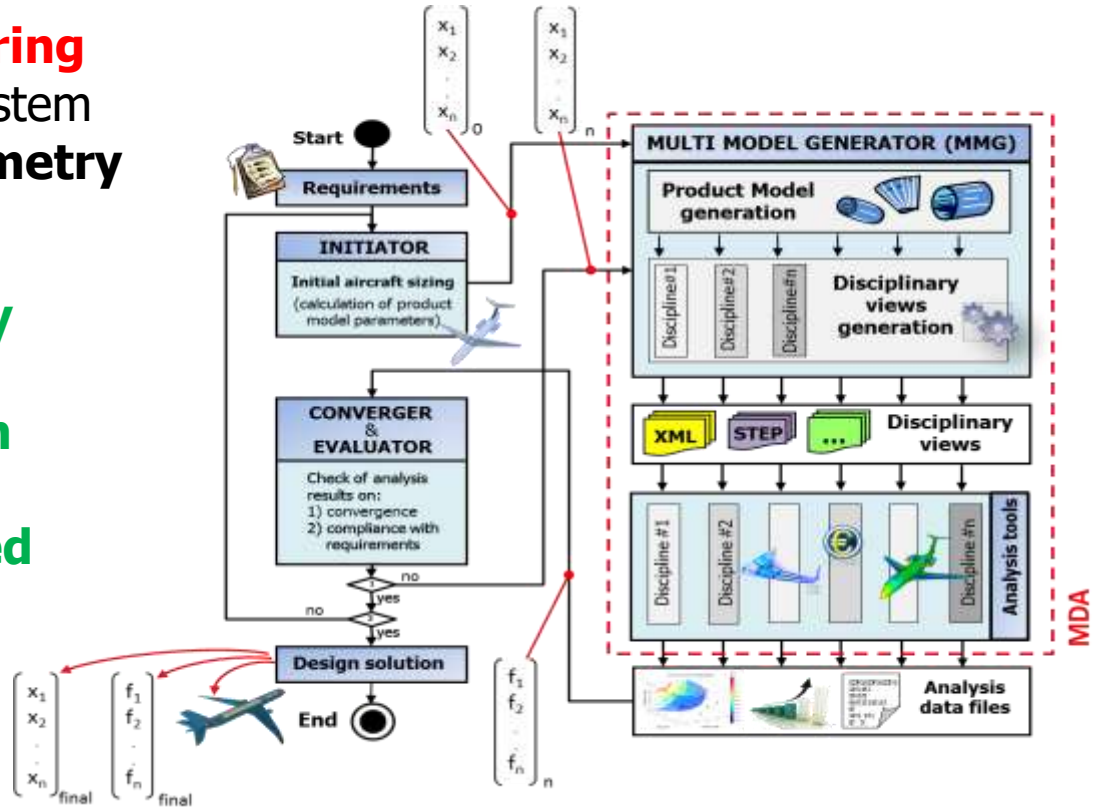


Automatically generated discipline-specific views

MDO and MDO implementation systems

The Design and Engineering Engine (DEE): an MDO system based on **generative geometry approach**

- Truly multidisciplinary
- Multi-fidelity
- Topology modification allowed
- Suitable for distributed design
- Suitable for novel design studies



The MMG functional components

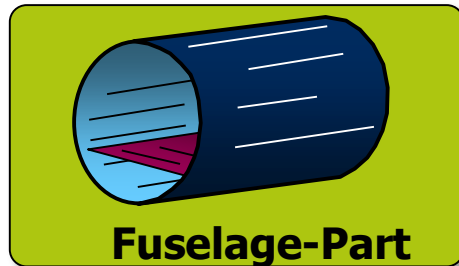
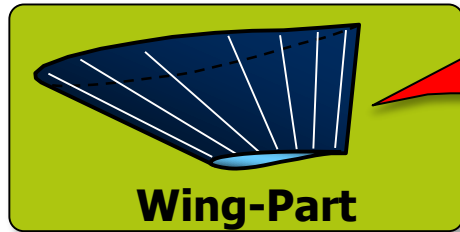
The 2 main constitutive elements of the aircraft Multi-Model Generator (MMG) are:

1. The **High-Level Primitives (HLPs)**: parametrical functional building blocks to model the geometry of different aircraft configurations
2. The **Capability Modules (CMs)**: operators (acting on the HLPs geometry) to automate the generation of dedicated models for multidisciplinary analysis



High Level Primitives for modular KBE product model development

HLPs for design space parameterization



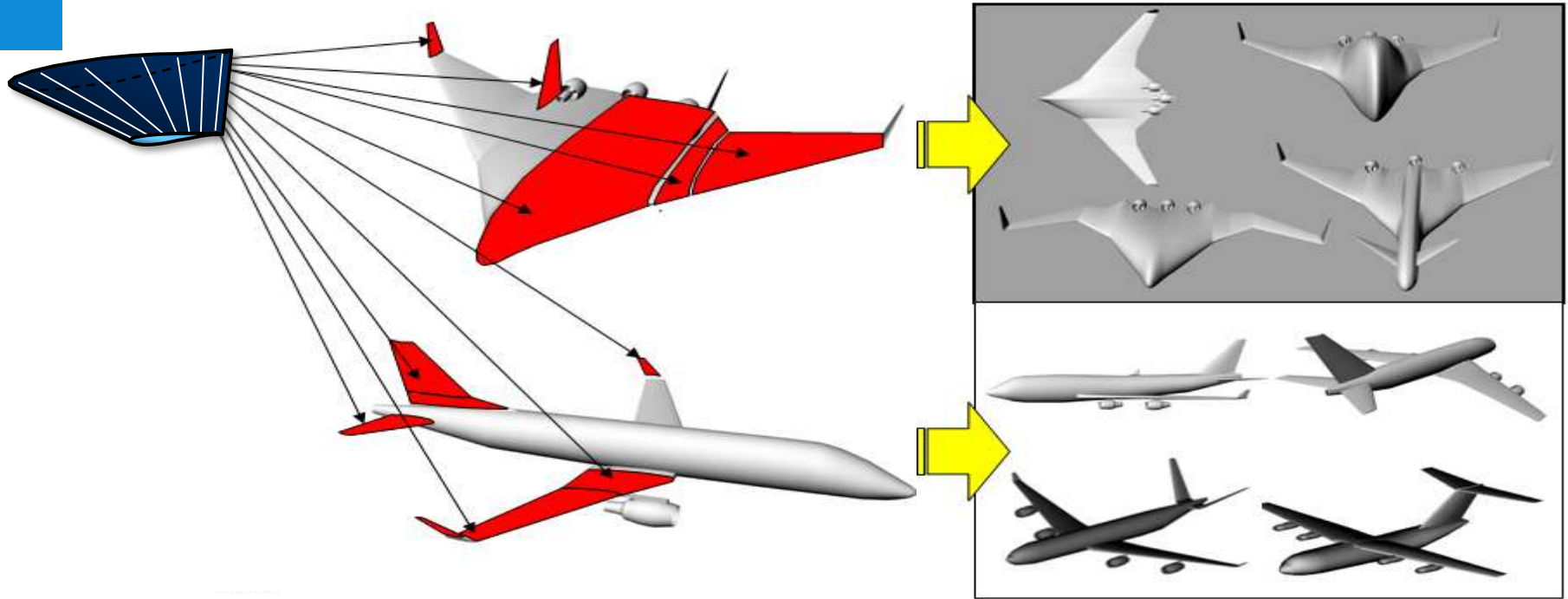
Wing-Part parameters

- Type of airfoil (from a library)
- Amount of airfoils
- Positioning of airfoils
- Thickness of airfoils
- Reference axes
- Chords' length
- Span
- Dihedral angle
- Sweep angle
- Twist angle
-

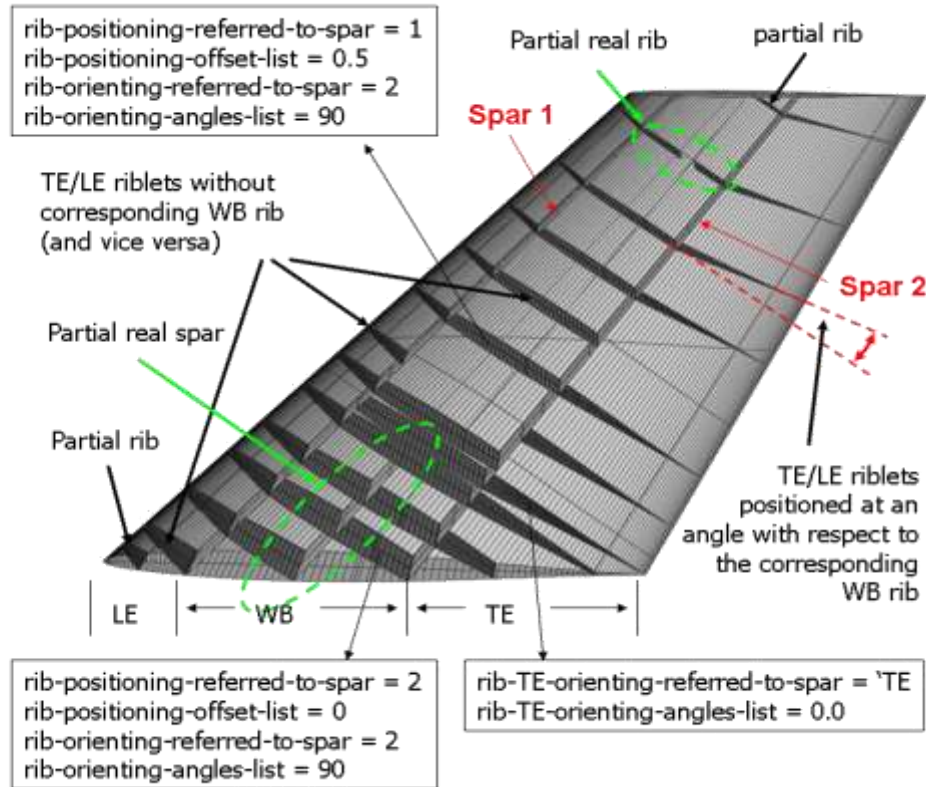


HLPs for design space parameterization

Enabled fast generation of **aircraft configurations** and **variants**



Structure definition in the HLPs



Number, location and orientation of each individual rib inside center wing-box, LE and TE sections defined by few parameters

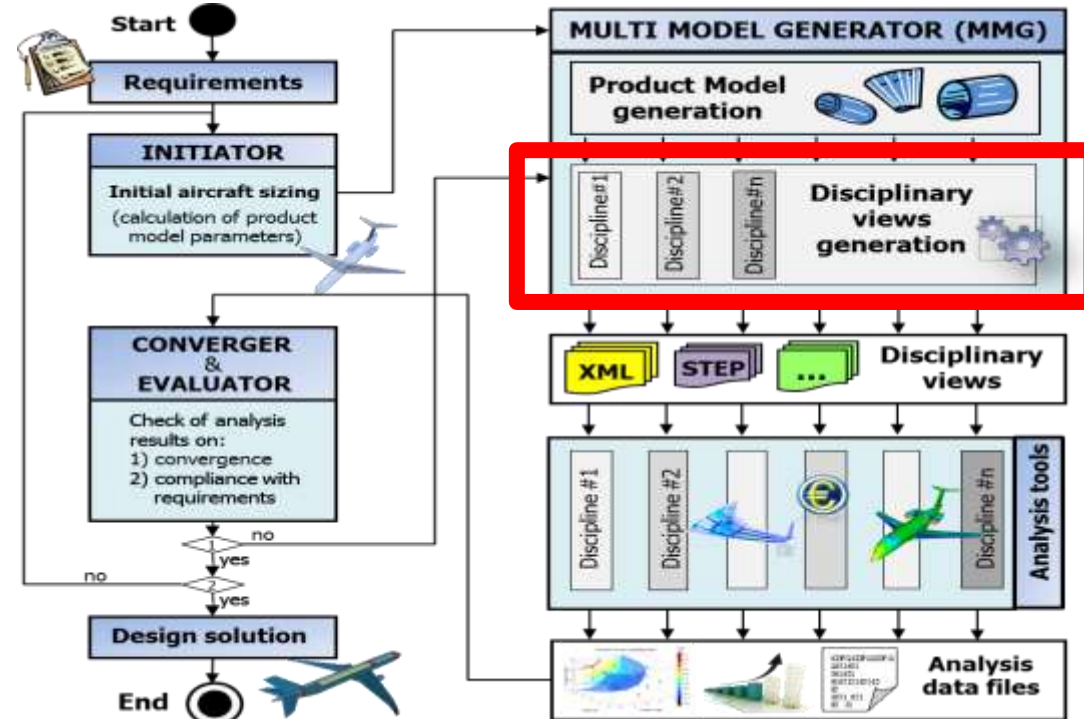
Capability modules to support multidisciplinary analysis

Capability Modules (CM) for disciplinary views generation



On how different specialists look at the same product and need different product representations (*views*)

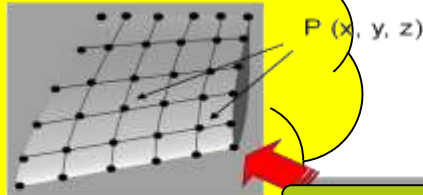
Capability Modules (CM) for disciplinary views generation



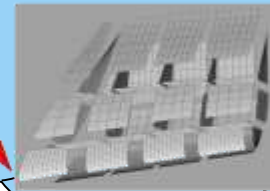
CM for disciplinary views generation



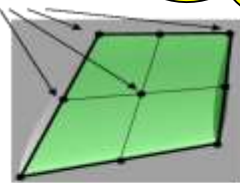
Generation of cloud of point for CFD analysis



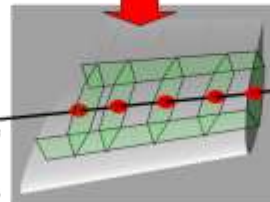
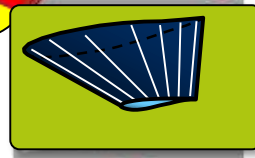
Generation of meshable elements for FE analysis



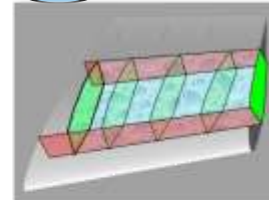
P (x, y, z)



Generation of flat panels



Generation of condensation axis and masses



Fuel Tank volume calculation

Aerodynamics office

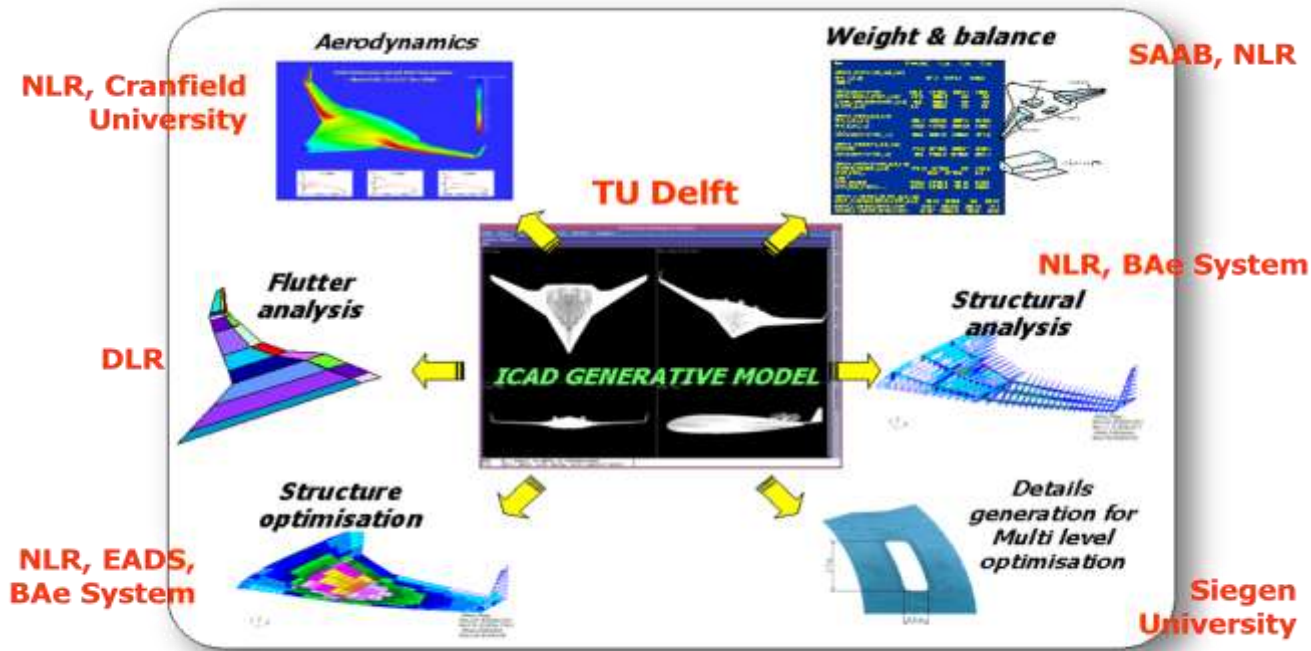
Stress office

*The first collaborative MDO
system for aircraft design,
supported by KBE*

MOB project (2000)

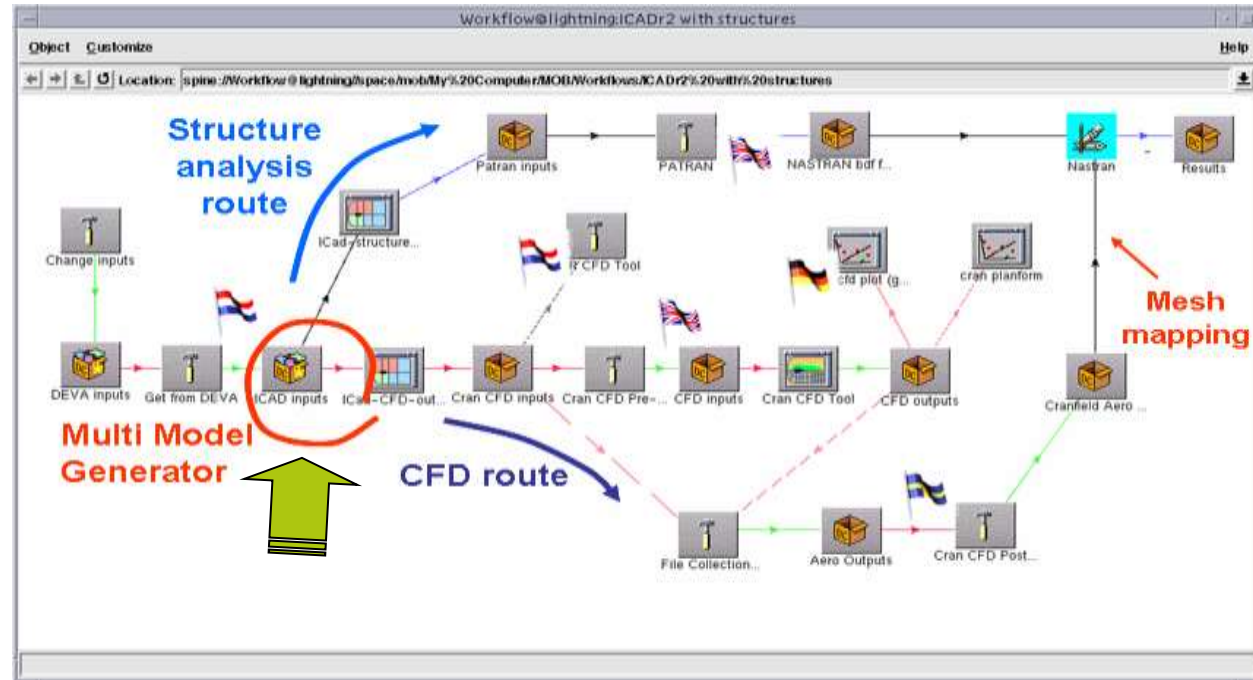
The EU MOB project study case

Development of a collaborative computational design system for MDO of blended wing body aircraft



The EU MOB project study case

Development of a collaborative computational design system for MDO of blended wing body aircraft



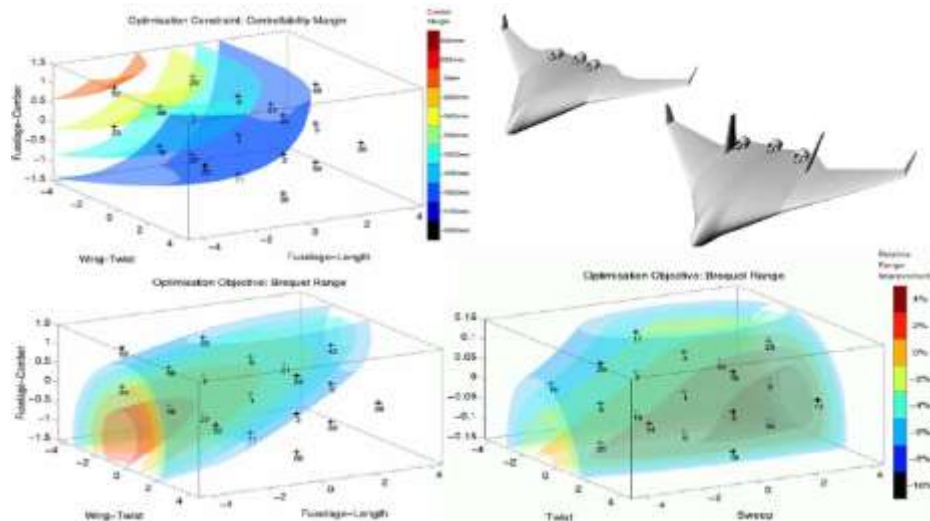
The MOB distributed design process integrated by means of a PIDO* tool

The EU MOB project study case

Development of a collaborative computational design system for MDO of blended wing body aircraft

Automatic generation and analysis of **>50 different BWB configurations** in **2 days**, fully hands-off.

Results presented via *response surfaces*



Disciplines involved:

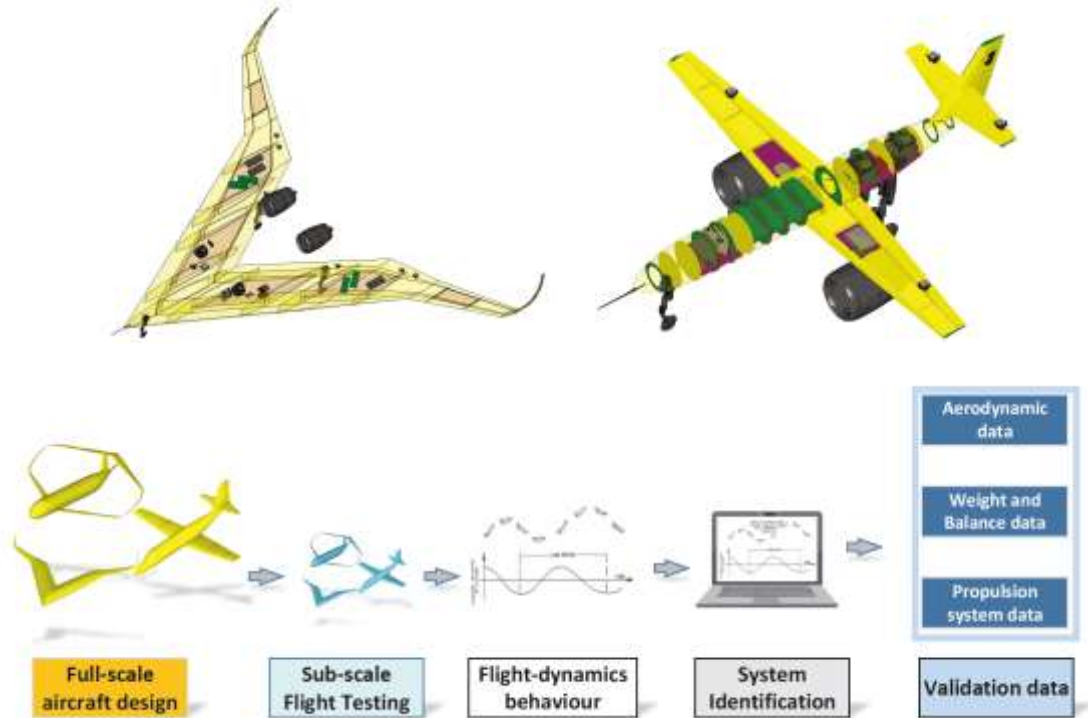
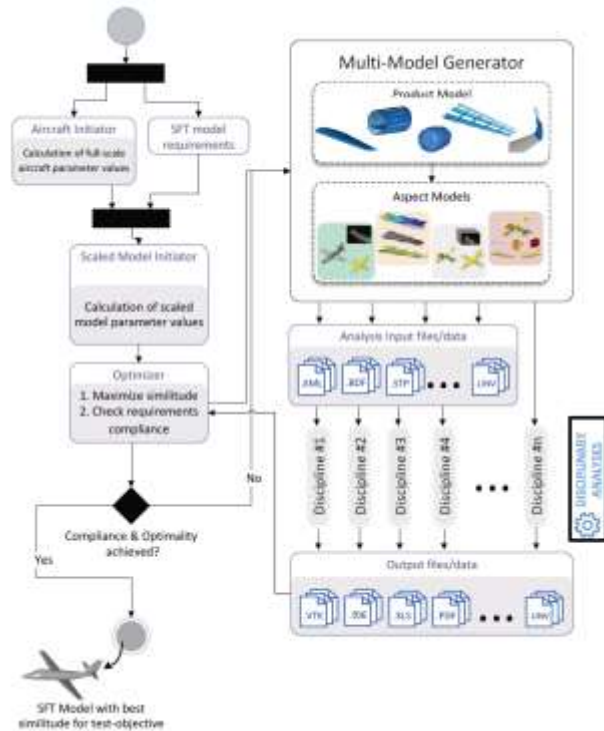
- Weight&balance
- Aerodynamics (CFD-Euler)
- Structure (full blown FEM with internal optimisation loop for sizing)
- Aeroelastics
- Flight mechanics

The DEE for scaled flight test model design (2022)

*An MDAO-based similarity maximization
approach to support SFT*

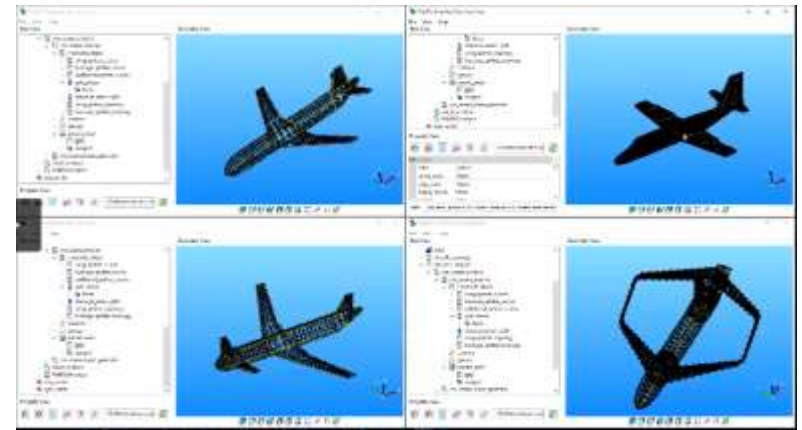
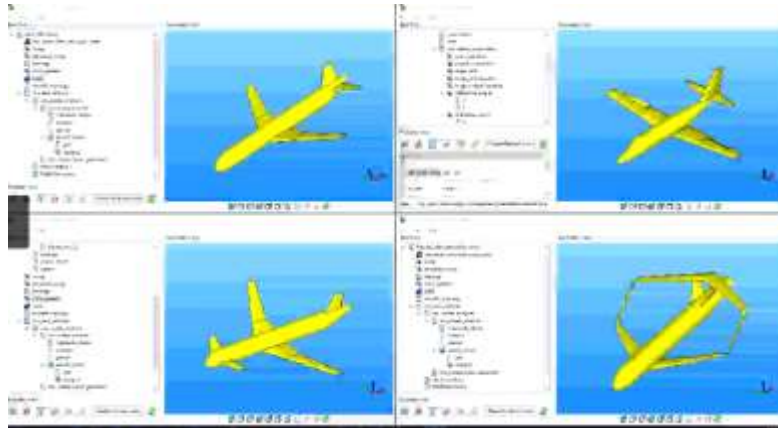
The DEE for scaled flight test model design

An MDAO-based similarity maximization approach to support SFT



The aircraft Multi Model Generator

Check MMG videoclips on Brightspace

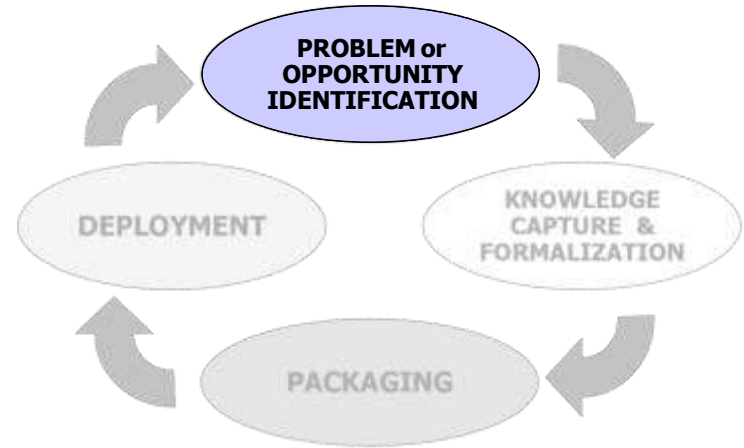


Development methodology:

The KBE app “life cycle”

KBE application lifecycle: *problem identification*

- Identification of a **problem/opportunity** to develop a new KBE application*
- Identification of:
 - **knowledge sources** (e.g., domain experts, manuals, literature, data-bases, etc.)
 - **knowledge acquisition** tools and techniques for **knowledge representation**.

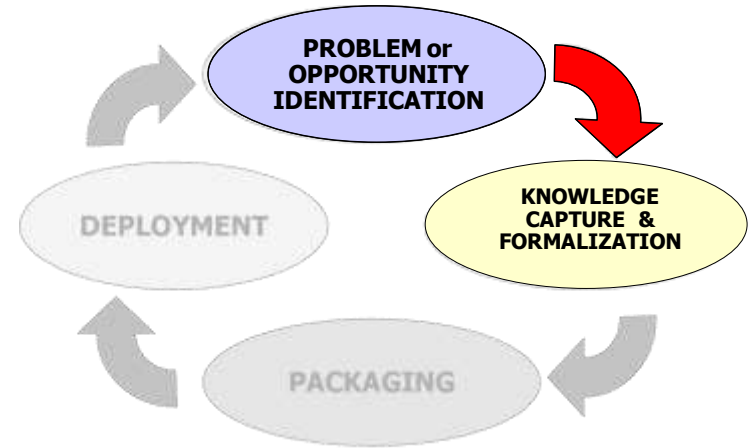


If the expected benefit and technical feasibility justify the development case, management can give approval to proceed to the next lifecycle phase.

KBE application lifecycle: *Capture&Formalize*

This phase includes two main activities:

- **K. Capture:** engineering knowledge is extracted from the identified sources and structured through *informal representations based on natural language* (e.g. diagrams, sketches, illustrations, text.)*.
- **K. Formalization:** the captured knowledge is refined and reshaped into *formal - yet programming language neutral – representations suitable to initiate code development* (e.g. UML diagrams and algorithms pseudo-code)**.



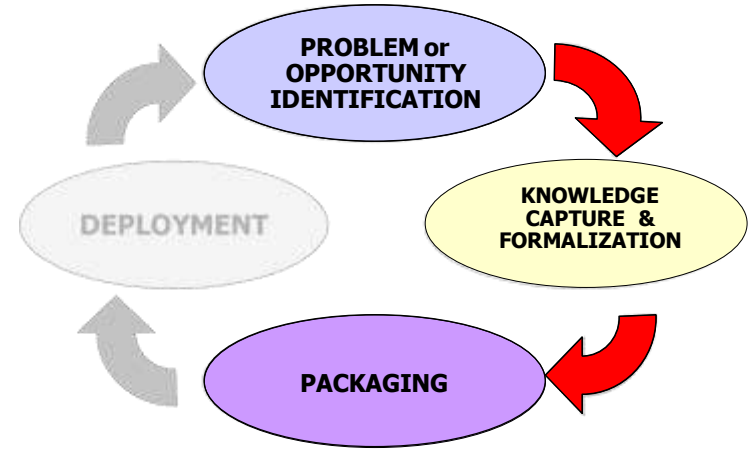
No coding on the KBE platform has actually started in this phase.

KBE application lifecycle: *Packaging**

In this phase the knowledge formalized in the previous phase **is coded into the selected KBE environment** (e.g. ParaPy).

This is the actual programming phase in the KBE lifecycle

This phase sets the **transition*** from the **neutral**, KBE system-independent knowledge representation, **to the specific** format dictated by the language of the selected KBE system



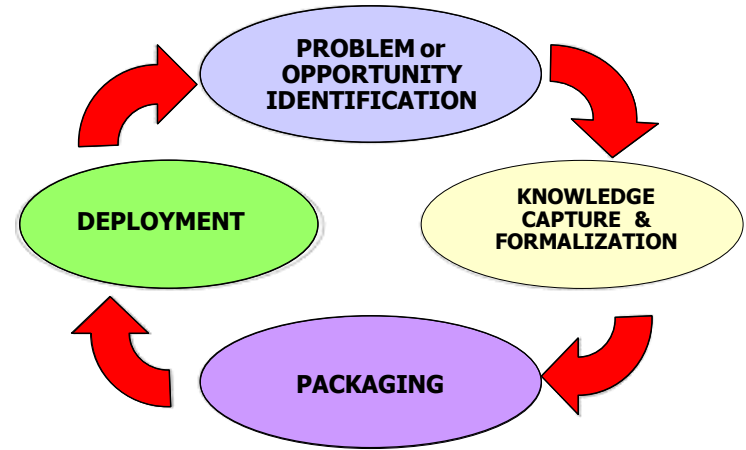
**implementation*

KBE application lifecycle: *Deployment*

In this phase the application is distributed to the potential users.

The KBE tool is installed, **documentation is provided*** and, eventually, end-users are trained.

After deployment, new opportunities to improve or extend the application might become evident **and a new cycle can start**



Your KBE app “life cycle” in this course

Your KBE assignment

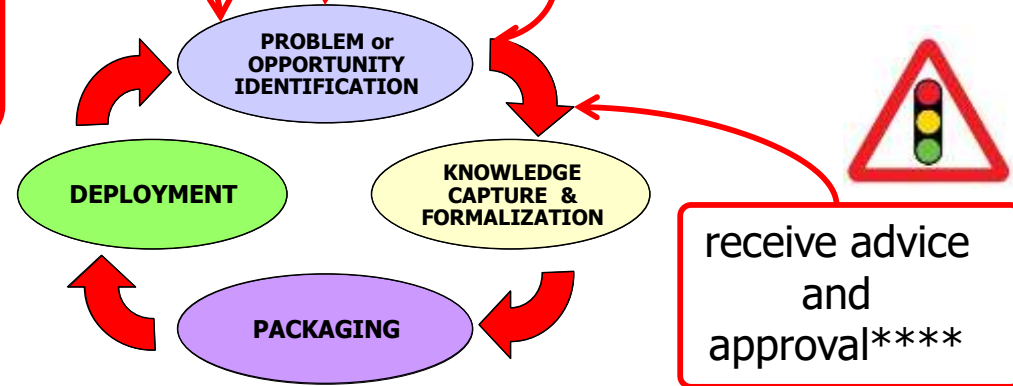
Identify an engineering problem that is:

- **suitable** for KBE application
- **feasible** within...
 - given time
 - available engineering knowledge
 - capability/limitations of the given KBE platform
 - its licensing scheme

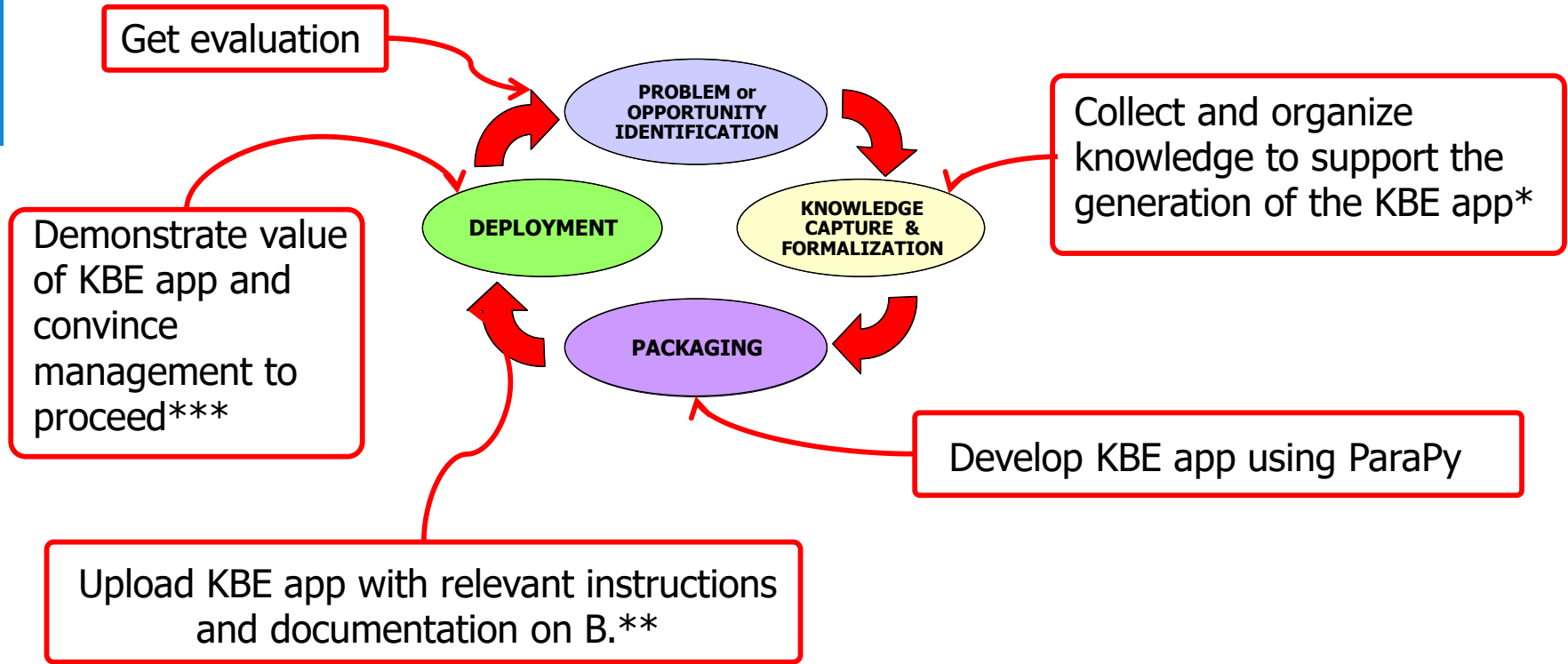
Identify:

- knowledge sources/tools*
- knowledge representation means**

Deliver assignment description***



Your KBE assignment



AE4205 (MDO) and AE4204 (KBE) in 3 bullet points

The MDO and KBE courses in 3 bullet points

1. Innovative aircraft configurations are needed to meet future aviation requirements. However, conventional design methods are limited and not (yet) adequate to address the design of novel and unconventional concepts aircraft
2. Multidisciplinary Design Optimization is required
3. KBE technology enables MDO by:
 - parameterizing complex products
 - automating model generation and analysis workflows
 - integrating multiple disciplines

Main reference material for this lecture

1. **Book** *MDO supported by KBE* by Wiley: **Chapter 9 on KBE**
2. **Journal paper** *Knowledge based engineering: Between AI and CAD. Review of a language based technology to support engineering design*,
G. La Rocca,, Journal of Advanced Engineering Informatics, Vol. 26, pp.159–179, 2012.
3. **CHAPTER 7** *Knowledge Based Engineering. Opportunities and Methodology*
4. **CHAPTER 4** *Conceptual development of the MMG . High Level Primitives and Capability Modules*
5. **CHAPTER 5** *Implementation of the High-Level Primitive concept in the KBE system*
6. **Videoclips on Multi Model Generator: KBE to enable MDO**

Ref 2-5 available on Brightspace, *Readings material* section

